

OSPF AREA

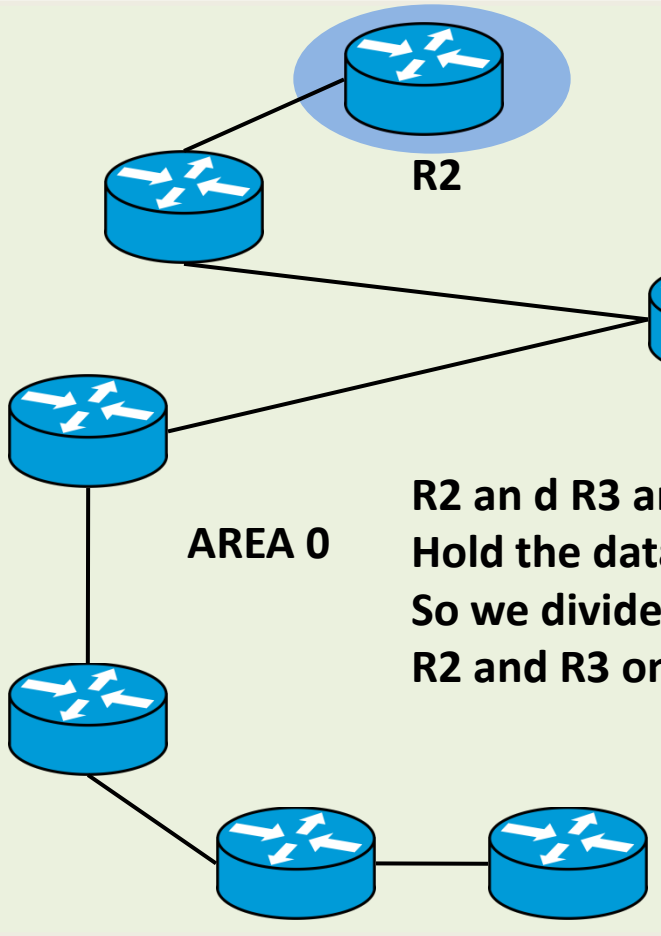
1. What is an OSPF Area?

An OSPF area is a logical grouping of routers that share the same link-state information. Areas are used to reduce the amount of routing information that needs to be processed by each router, thereby optimizing the network's performance. By segmenting the network into areas, OSPF limits the propagation of routing updates, reducing the amount of processing required by each router.

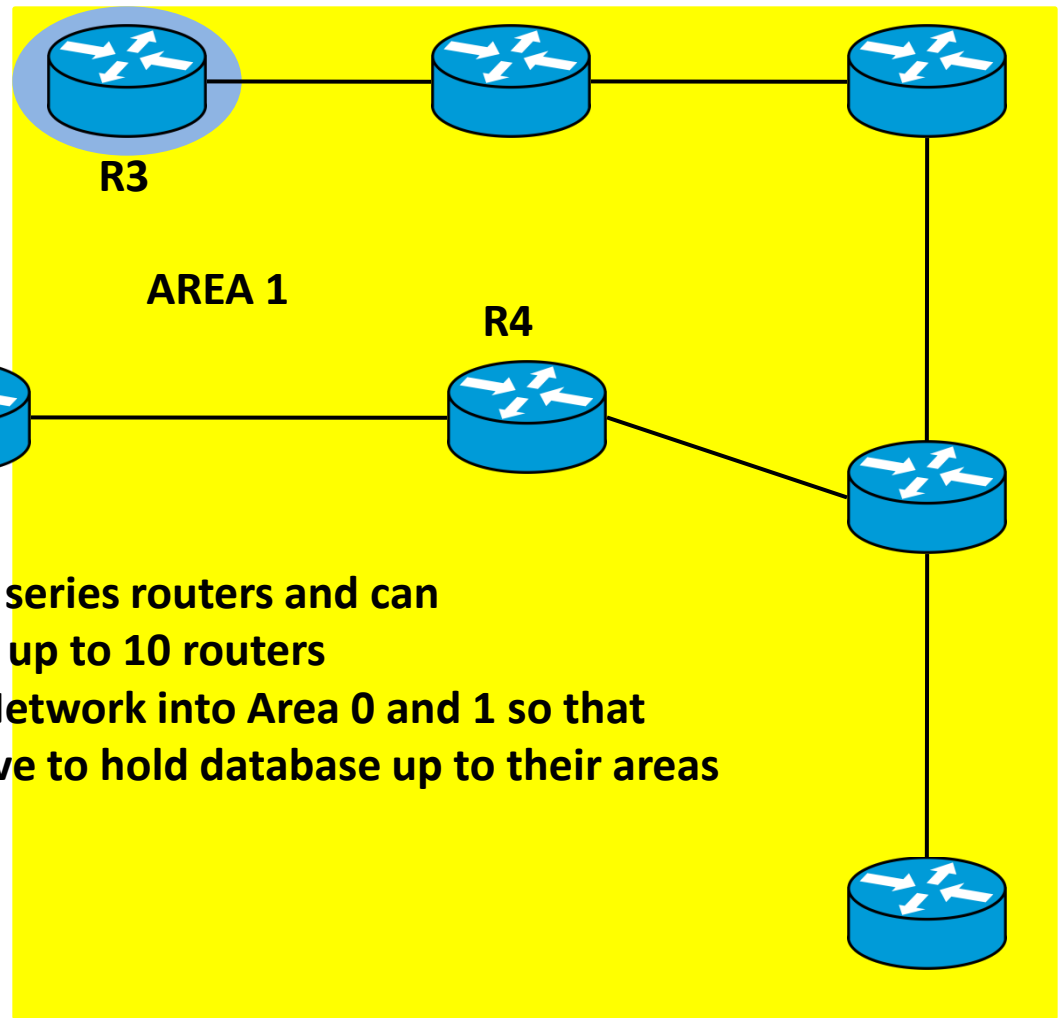
2. Types of OSPF Areas

1. Backbone Area (Area 0)
2. Standard Area

Why OSPF Area:



R2 and R3 are old series routers and can
Hold the database up to 10 routers
So we divide the Network into Area 0 and 1 so that
R2 and R3 only have to hold database up to their areas



TYPES of OSPF AREAS:

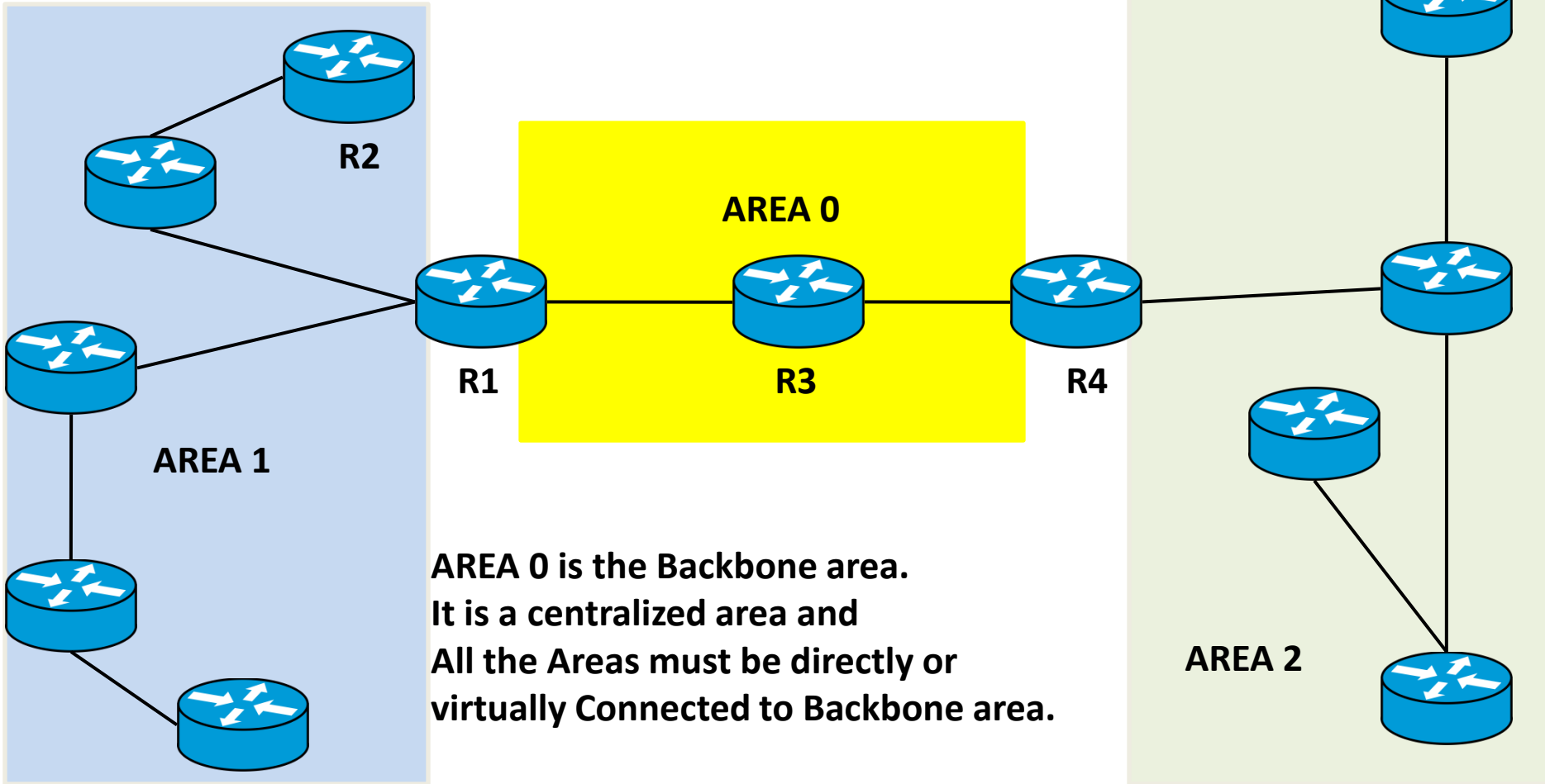
1. Backbone Area (Area 0)
2. Standard Area
3. Stub Area
4. Totally Stubby Area
5. Not-So-Stubby Area (NSSA)
6. Totally NSSA

OSPF AREA Types

Backbone Area (Area 0): This is the core area of an OSPF network. All other areas must connect to the backbone area directly or through a virtual link. The backbone area serves as the central hub for OSPF routing and is responsible for inter-area routing.

Standard Area: A standard area is a non-backbone area that can connect to Area 0. It contains routers and networks that exchange routing information only within the area. In standard areas, all routers maintain a complete link-state database of the area.

Backbone Area: Area 0



Advantages of Using OSPF Areas:-

Scalability: By dividing a large network into smaller areas, OSPF reduces the amount of routing information that each router must process, improving scalability.

Improved Performance: Limiting the scope of routing updates reduces CPU and memory usage on routers.

Faster Convergence: Smaller OSPF areas allow for quicker detection and correction of network changes, improving convergence times.

Reduced Overhead: Using stub and totally stubby areas reduces the size of the routing table, which helps in lowering the processing overhead.

Design Considerations for OSPF Areas

Area Size: Keep areas small to maintain fast convergence and minimize the size of the routing table. Large areas can lead to performance degradation.

Connectivity to Area 0: All non-backbone areas must connect to the backbone (Area 0) either directly or through virtual links.

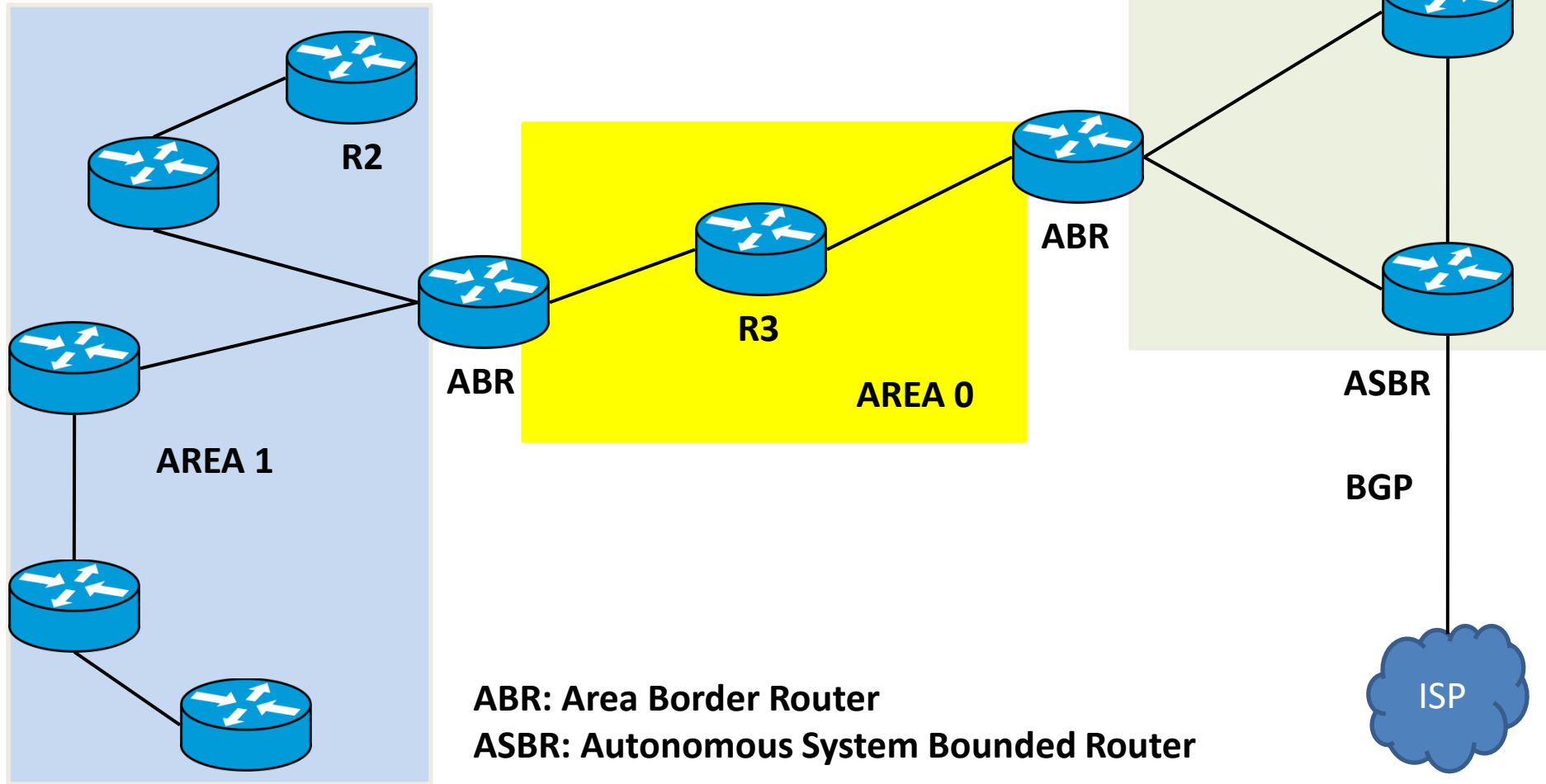
Use of Stub Areas: Use stub or totally stubby areas for regions where no external routing is needed, such as smaller branches or remote sites.

Area Border Routers (ABRs) and Autonomous System Boundary Routers (ASBRs)

Area Border Routers (ABRs): These routers connect one or more OSPF areas to the backbone area (Area 0). ABRs are responsible for summarizing and redistributing routing information between areas.

Autonomous System Boundary Routers (ASBRs): These routers connect an OSPF network to other routing domains or external networks. ASBRs are responsible for importing and exporting routing information between OSPF and non-OSPF networks.

ABR and ASBR: -



OSPF Configuration:

Router>en

Router#conf t

Router(config)#router ospf 1

Router(config-router)#network 200.0.0.0 0.0.0.255 area 0

Database Table/Brief Version of Topology Table:

Router#Show ip ospf database

ASBR Summary/Type 4 LSA:

Router#Show ip ospf database asbr-summary

OSPF Configuration:

Display cost for each interface:

Router#Show ip ospf interface brief

Best Cost for reaching the Border Routers from internal:

Router#Show ip ospf border-routers

Type 7 LSA:

Router#Show ip ospf database nssa-external

Display External Route OSPF:

Router#Show ip route ospf

OSPF Commands:

Type 5 LSA or external route to OSPF:

Router#Show ip ospf database external

Best Cost for reaching the ASBR from internal:

Router#Show ip ospf database asbr-summary